

UNION TOOL

Tungsten Carbide End Mills UNIMAX Series

NEW

HARDMAX 3 Flutes Taper Neck Ball End Mills

HFTNB

Total 73 Models



UNION TOOL CO.

HARDMAX 3 Flutes Taper Neck Ball End Mills

Size **R0.5~R2**



HFTNB

Super
MG

HARD
MAX

40°

R
±0.005
R0.5~R1.5

R
±0.007
R2

Shank Dia
0/-0.005

Back Taper
Geometry

Variable
Pitch

NEW

Material Applications (☆ Highly Recommended ◎ Recommended ○ Suggested)

Work Material															
CARBON STEELS S45C S55C	ALLOY STEELS SK / SCM SUS	PRE-HARDENED STEELS NAK HPM	HARDENED STEELS			CAST IRON	ALUMINUM ALLOYS	GRAPHITE	COPPER	PLASTICS	GLASS FILLED PLASTICS	TITANIUM ALLOYS	HEAT RESISTANT ALLOYS	CEMENTED CARBIDE	HARD BRITTLE (NON-METALLIC) MATERIALS
			~ 55HRC	~ 60HRC	~ 70HRC										
					 ~ 65HRC										

Features

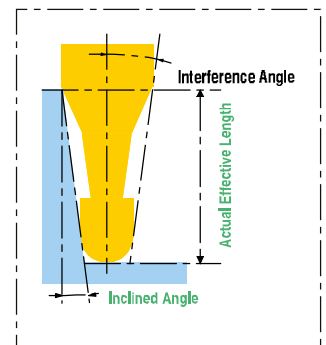
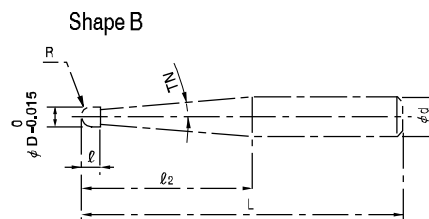
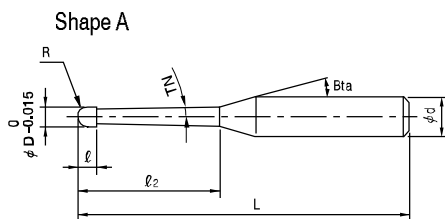
3 Flute Taper Neck Ball End Mills for Hard Materials.

The negative rake angle design improves wear resistance.

Back taper design reduces cutting resistance.

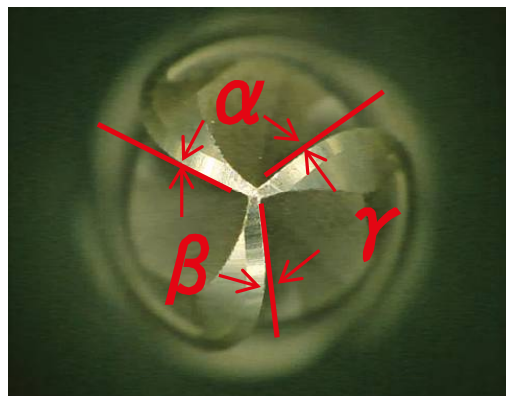
Suitable for both roughing and finishing. Diameter Tolerance : 0/-0.015.

The shank taper angle shown is not an exact value and to avoid contact with the workpiece, we recommend the user controls the precise value of this angle. Shank taper angle should not make contact with the work piece.



* Variable Pitch design

Minimizing vibration and chattering



※ Variable Pitch
 $\alpha \neq \beta \neq \gamma$

* A wide choice of Taper Neck Angles

Useful sizes available: 0.4° · 0.9° · 1.4° · 1.9° · 2.9°

Using with HTNRS, Taper Neck Radius End Mill, offers higher efficient milling.

HARDMAX 3 Flutes Taper Neck Ball End Mills

Total 73 models

Unit (mm)

Model Number	Radius of Ball Nose R	Neck Taper Angle TN	Neck Length ℓ_2	Length of Cut ℓ	Shank Taper Angle Bta	Overall Length L	Shank Diameter ϕd	Shape	Price ¥	Interference Angle	Effective Length by Inclined Angles — : Interference				
											30°	1°	1°30'	2°	3°
HFTNB 3010-080-08	R0.5	0.4°	8	0.8	16°	60	6	A	8,690	8.82°	8.05	8.30	8.57	8.85	9.48
HFTNB 3010-100-08			10			60	6		8,690	7.89°	10.07	10.38	10.71	11.07	11.87
HFTNB 3010-120-08			12			60	6		8,690	7.13°	12.08	12.46	12.86	13.29	14.26
HFTNB 3010-160-08			16			60	6		8,690	5.98°	16.10	16.61	17.16	17.74	19.03
HFTNB 3010-200-08			20			60	6		11,040	5.15°	20.13	20.77	21.45	22.18	23.81
HFTNB 3010-260-08			26			70	6		11,590	4.26°	26.17	27.00	27.89	28.85	30.97
HFTNB 3010-300-08			30			70	6		12,140	3.82°	30.19	31.16	32.19	33.29	35.75
HFTNB 3010-060-18		0.9°	6			60	6		8,690	10.11°	-	6.06	6.25	6.45	6.90
HFTNB 3010-080-18			8			60	6		8,690	8.94°	-	8.07	8.33	8.60	9.21
HFTNB 3010-100-18			10			60	6		8,690	8.01°	-	10.08	10.41	10.75	11.53
HFTNB 3010-120-18			12			60	6		8,690	7.25°	-	12.10	12.49	12.91	13.84
HFTNB 3010-160-18			16			60	6		8,690	6.10°	-	16.12	16.65	17.21	18.47
HFTNB 3010-200-18			20			60	6		11,040	5.26°	-	20.15	20.81	21.52	23.09
HFTNB 3010-260-18			26			70	6		11,590	4.36°	-	26.19	27.05	27.98	30.03
HFTNB 3010-300-18			30			70	6		12,140	3.91°	-	30.21	31.21	32.28	34.66
HFTNB 3010-100-28		1.4°	10			60	6		8,690	8.13°	-	-	10.10	10.43	11.18
HFTNB 3010-120-28			12			60	6		8,690	7.38°	-	-	12.11	12.52	13.42
HFTNB 3010-160-28			16			60	6		8,690	6.22°	-	-	16.14	16.68	17.90
HFTNB 3010-200-28			20			60	6		11,040	5.37°	-	-	20.17	20.85	22.38
HFTNB 3010-260-28			26			70	6		11,590	4.46°	-	-	26.21	27.10	29.09
HFTNB 3015-100-08	R0.75	0.4°	10	1.2	16°	60	6	A	9,110	7.55°	10.11	10.41	10.74	11.09	11.87
HFTNB 3015-160-08			16			60	6		9,940	5.64°	16.14	16.65	17.18	17.75	19.03
HFTNB 3015-200-08			20			60	6		9,940	4.82°	20.17	20.80	21.48	22.20	23.81
HFTNB 3015-300-08			30			70	6		10,760	3.54°	30.23	31.19	32.21	33.31	35.75
HFTNB 3015-100-18		0.9°	10			60	6		9,110	7.66°	-	10.13	10.45	10.79	11.54
HFTNB 3015-160-18			16			60	6		9,940	5.75°	-	16.17	16.69	17.24	18.48
HFTNB 3015-200-18			20			60	6		9,940	4.93°	-	20.19	20.85	21.55	23.11
HFTNB 3015-300-18			30			70	6		10,760	3.63°	-	30.26	31.25	32.31	34.67
HFTNB 3015-100-28		1.4°	10			60	6		9,110	7.79°	-	-	10.15	10.48	11.21
HFTNB 3015-160-28			16			60	6		9,940	5.87°	-	-	16.19	16.73	17.93
HFTNB 3015-200-28			20			60	6		9,940	5.03°	-	-	20.22	20.90	22.41
HFTNB 3015-300-28			30			80	6		10,760	3.72°	-	-	30.28	31.31	33.60
HFTNB 3020-120-08	R1	0.4°	12	1.6	16°	60	6	A	9,100	6.40°	12.12	12.49	12.87	13.29	14.22
HFTNB 3020-160-08			16			60	6		9,320	5.27°	16.15	16.64	17.17	17.73	18.99
HFTNB 3020-200-08			20			60	6		9,940	4.47°	20.17	20.80	21.46	22.18	23.77
HFTNB 3020-220-08			22			70	6		9,940	4.16°	22.18	22.87	23.61	24.40	26.16
HFTNB 3020-260-08			26			70	6		11,040	3.65°	26.21	27.03	27.90	28.84	30.93
HFTNB 3020-300-08			30			70	6		12,140	3.25°	30.23	31.18	32.20	33.29	35.71
HFTNB 3020-320-08			32			80	6		12,140	3.08°	32.25	33.26	34.35	35.51	38.09
HFTNB 3020-400-08			40			80	6		14,350	2.55°	40.30	41.57	42.94	44.40	No Interference
HFTNB 3020-120-18		0.9°	12			60	6		9,100	6.52°	-	12.15	12.52	12.93	13.83
HFTNB 3020-160-18			16			60	6		9,320	5.38°	-	16.17	16.68	17.23	18.46
HFTNB 3020-200-18			20			60	6		9,940	4.57°	-	20.20	20.85	21.54	23.08
HFTNB 3020-260-18			26			70	6		11,040	3.74°	-	26.24	27.09	28.00	30.02
HFTNB 3020-300-18			30			70	6		12,140	3.33°	-	30.26	31.25	32.30	34.65
HFTNB 3020-360-18			36			80	6		12,140	2.86°	-	36.30	37.49	38.76	No Interference
HFTNB 3020-400-18			40			80	6		14,350	2.62°	-	40.33	41.65	43.06	No Interference
HFTNB 3020-500-18			50			100	6		16,000	2.16°	-	50.39	52.05	53.83	No Interference
HFTNB 3020-160-28		1.4°	16			60	6		9,320	5.49°	-	-	16.20	16.73	17.92
HFTNB 3020-200-28			20			60	6		9,940	4.68°	-	-	20.23	20.90	22.40
HFTNB 3020-260-28			26			70	6		11,040	3.83°	-	-	26.27	27.15	29.11
HFTNB 3020-300-28			30			70	6		12,140	3.41°	-	-	30.30	31.32	33.59
HFTNB 3020-400-28			40			80	6		15,070	2.69°	-	-	40.36	41.73	No Interference
HFTNB 3020-620-38		1.9°	62			100	6	B	18,000	-	-	-	-	No Interference	No Interference
HFTNB 3020-410-58		2.9°	41			80	6		17,000	-	-	-	-	-	No Interference

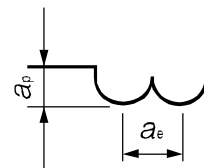
HARDMAX 3 Flutes Taper Neck Ball End Mills

Model Number	Radius of Ball Nose R	Neck Taper Angle TN	Neck Length ℓ_2	Length of Cut ℓ	Shank Taper Angle Bta	Overall Length L	Shank Diameter ϕd	Shape	Price ¥	Interference Angle	Effective Length by Inclined Angles — : Interference				
											30°	1°	1°30'	2°	3°
HFTNB 3030-200-08	R1.5	0.4°	20	2.4	16°	60	6	A	11,200	3.67°	20.23	20.84	21.49	22.19	23.75
HFTNB 3030-260-08			26			70	6		11,960	2.94°	26.27	27.07	27.93	28.86	No Interference
HFTNB 3030-300-08			30			70	6		13,660	2.60°	30.29	31.23	32.23	33.30	No Interference
HFTNB 3030-320-08			32			80	6		14,350	2.46°	32.31	33.31	34.38	35.52	No Interference
HFTNB 3030-400-08			40			80	6		15,180	2.01°	40.36	41.62	42.97	44.41	No Interference
HFTNB 3030-200-18		0.9°	20			60	6		11,200	3.76°	-	20.27	20.90	21.58	23.09
HFTNB 3030-300-18			30			70	6		13,660	2.67°	-	30.34	31.31	32.34	No Interference
HFTNB 3030-400-18			40			80	6		15,180	2.07°	-	40.40	41.71	43.11	No Interference
HFTNB 3030-500-18			50			100	6		17,500	1.69°	-	50.46	52.11	No Interference	No Interference
HFTNB 3030-600-18			60			100	6		19,000	1.43°	-	60.52	No Interference	No Interference	No Interference
HFTNB 3030-400-28		1.4°	40			80	6		15,180	2.12°	-	-	40.45	41.80	No Interference
HFTNB 3030-500-28			50			100	6		17,000	1.74°	-	-	50.51	No Interference	No Interference
HFTNB 3030-650-28			65			100	6	B	19,000	-	-	-	No Interference	No Interference	No Interference
HFTNB 3040-300-18	R2	0.9°	30	6	16°	80	6	A	13,330	1.90°	-	30.51	31.47	No Interference	No Interference
HFTNB 3040-400-18			40			80	6		17,250	1.45°	-	40.58	No Interference	No Interference	No Interference
HFTNB 3040-500-18			50			100	6		19,000	1.18°	-	50.64	No Interference	No Interference	No Interference
HFTNB 3040-600-18			60			100	6		20,200	0.99°	-	No Interference	No Interference	No Interference	No Interference
HFTNB 3040-480-28		1.4°	48		-	80	6	B	17,250	-	-	-	No Interference	No Interference	No Interference

WORK MATERIAL			PREHARDENED STEELS/HARDENED STEELS NAK / SKD (30~45HRC)				HARDENED STEELS SKD / SKT (45~55HRC)				HARDENED STEELS SKD / SKH (55~65HRC)			
Model Number	Radius of Ball Nose (mm)	Neck Length (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a_p Axial Depth (mm)	a_e Radial Depth (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a_p Axial Depth (mm)	a_e Radial Depth (mm)	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	a_p Axial Depth (mm)	a_e Radial Depth (mm)
3010	R0.5	6	14,500	1,300	0.1	0.2	14,500	1,250	0.06	0.12	14,500	1,200	0.04	0.08
		8	14,000	1,200	0.09	0.18	13,750	1,160	0.06	0.1	13,500	1,120	0.04	0.06
		10	13,300	1,000	0.08	0.16	12,650	1,000	0.05	0.09	12,000	1,000	0.04	0.05
		12	13,000	870	0.07	0.14	12,000	850	0.04	0.08	11,000	880	0.03	0.05
		16	12,500	680	0.05	0.1	10,250	600	0.04	0.06	8,000	550	0.03	0.04
		20	12,000	600	0.04	0.08	9,500	500	0.03	0.06	7,000	400	0.02	0.04
		26	11,700	520	0.03	0.06	8,600	370	0.02	0.04	5,500	220	0.02	0.03
		30	11,500	500	0.02	0.05	8,250	350	0.02	0.04	5,000	200	0.02	0.03
3015	R0.75	10	12,000	1,230	0.13	0.3	11,500	1,100	0.09	0.2	11,000	1,100	0.06	0.14
		16	11,200	930	0.1	0.25	10,600	910	0.07	0.16	10,000	900	0.05	0.11
		20	10,800	750	0.08	0.22	9,500	700	0.06	0.14	8,200	680	0.04	0.09
		30	10,000	550	0.06	0.16	8,300	450	0.04	0.1	6,600	380	0.03	0.08
3020	R1	12	10,300	1,200	0.16	0.38	10,150	1,130	0.12	0.25	10,000	1,100	0.1	0.18
		16	10,000	1,100	0.15	0.35	9,900	1,100	0.1	0.23	9,800	1,050	0.09	0.16
		20	9,500	950	0.15	0.32	9,300	940	0.1	0.21	9,000	930	0.08	0.15
		22	9,400	900	0.14	0.3	9,100	850	0.09	0.2	8,600	840	0.08	0.14
		26	9,300	750	0.12	0.28	8,700	730	0.08	0.2	8,000	700	0.07	0.13
		30	9,200	630	0.11	0.25	8,400	590	0.08	0.17	7,500	550	0.05	0.1
		32	8,800	580	0.1	0.24	8,200	550	0.07	0.16	7,300	480	0.04	0.1
		36	8,700	570	0.09	0.22	7,900	510	0.07	0.16	7,000	450	0.05	0.1
		40	8,300	500	0.08	0.2	7,500	450	0.06	0.15	6,600	400	0.04	0.1
		41	8,300	500	0.08	0.2	7,500	450	0.06	0.15	6,600	400	0.04	0.1
		50	8,000	430	0.06	0.15	6,700	340	0.04	0.12	5,300	250	0.03	0.1
		62	7,500	350	0.04	0.1	6,000	350	0.04	0.13	5,000	300	0.02	0.05
3030	R1.5	20	9,000	1,150	0.25	0.48	8,900	1,100	0.18	0.36	8,800	1,100	0.12	0.25
		26	8,600	1,000	0.22	0.42	8,300	1,000	0.16	0.32	8,200	980	0.11	0.22
		30	8,400	950	0.21	0.4	8,100	930	0.15	0.3	7,800	920	0.1	0.21
		32	8,300	900	0.2	0.37	7,800	860	0.14	0.28	7,400	840	0.09	0.2
		40	8,000	720	0.17	0.33	7,000	630	0.12	0.24	6,000	550	0.08	0.19
		50	7,600	570	0.14	0.28	6,400	450	0.09	0.2	5,200	400	0.06	0.17
		60	7,200	480	0.12	0.24	6,000	400	0.07	0.18	4,700	320	0.05	0.16
		65	7,200	480	0.12	0.24	6,000	400	0.07	0.18	4,700	320	0.05	0.16
3040	R2	30	8,000	1,100	0.35	0.55	7,800	1,050	0.24	0.4	7,600	1,000	0.16	0.33
		40	7,500	930	0.3	0.48	7,300	900	0.2	0.35	7,000	900	0.15	0.3
		48	7,200	750	0.26	0.42	6,500	650	0.16	0.3	5,800	600	0.13	0.27
		50	7,200	750	0.26	0.42	6,500	650	0.16	0.3	5,800	600	0.13	0.27
		60	7,000	600	0.22	0.36	6,000	520	0.13	0.26	5,000	440	0.11	0.25

Note:

- Decrease both spindle speed and feed rate proportionally when the milling parameters exceed the machine's maximum spindle speed.
- Decrease the feed rate more than 50% from the milling parameters when slot milling.
- Every coolant offers stable milling.



a_p : 軸方向の切込み深さ Axial Depth (mm)
 a_e : 半径方向の切込み深さ Radial Depth (mm)

Taper Pocket Milling Examples by Different Work Materials

NAK80 (40HRC)

HFTNB R1 × Neck Length 30 × Neck Taper Angle 0.9°

Milling Shape : Taper Pocket 30 x 5 x Depth 6 mm
Vertical Wall Inclined Angle 1°

After 102 min

HFTNB



No chipping after milling 3 pockets!

Competitor A

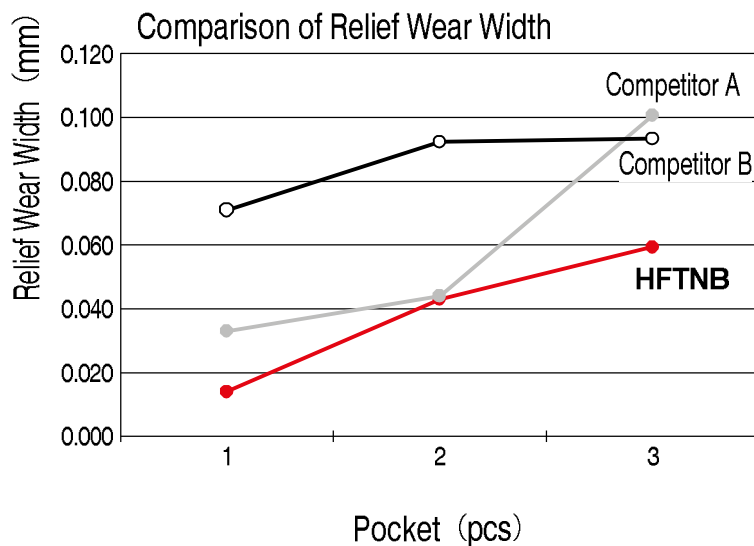


Chipping on the cutting edge after milling 3 pockets.

Competitor B



Chipping on the cutting edge after milling 1 pocket.



Relief Wear Width

(After milling 3 pockets)

Competitor A : 0.101 mm

Competitor B : 0.093 mm

HFTNB : 0.059 mm

Better relief wear than competitors.

Tool	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	Axial Depth a_p	Radial Depth a_e	Coolant	Cycle Time
HFTNB 3020-300-18	9,200 min ⁻¹	630 mm/min	0.11 mm	0.25 mm	Air Blow (Nozzle)	34 min / pocket (102 min / 3 pockets)

Taper Pocket Milling Examples by Different Work Materials

SKD61 (50HRC)

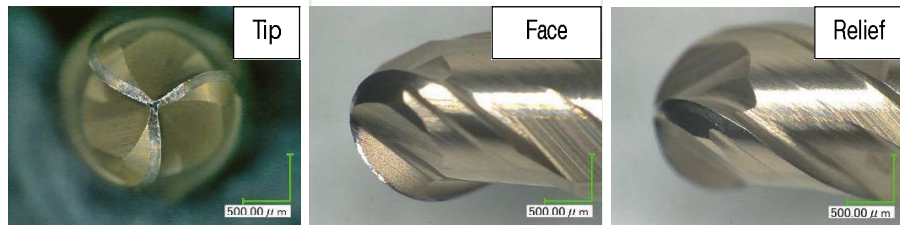
HFTNB R1 × Neck Length 30 × Neck Taper Angle 0.9°

Milling Shape : Taper Pocket 25 x 5 x Depth 4 mm
Vertical Wall Inclined Angle 1°

Tools after milling

HFTNB

After 40 min



After 120 min



**Normal wear condition after 120 min milling.
No chipping or any damages.**

Competitor A
After 40 min



Chipping on the cutting edges of all 3 flutes after the 1st pocket milling.

Competitor B
After 40 min



Large chipping on the cutting edge after the 1st pocket milling.

Chipping also on the other 2 flutes.

High durability under long cycle time.

Tool	Spindle Speed (min ⁻¹)	Feed Rate (mm/min)	Axial Depth a_p	Radial Depth a_e	Coolant	Cycle Time
HFTNB 3020-300-18	8,400 min ⁻¹	590 mm/min	0.08 mm	0.17 mm	Air Blow (Nozzle)	40 min / pocket



Advisory for Safe Use of UNIMAX Tungsten Carbide End Mills

Correct application and operation is strongly advised to avoid clogging, abrasion, etc, that could cause serious accidents or injuries. Ignition or sparks generated during milling could lead to fire or extreme damage to the work piece. End Mills are made with very sharp cutting edges and must be handled with extra care.

- * Never touch the cutting edge with your bare hands, as this could cause serious injury. Special caution is required when opening the package.
- * Dropping the tool could cause breakage or flying debris, leading to serious injury.
- * During milling, unexpected impact or shock on the tool could cause breakage or flying debris. Ensure to use protective items such as safety glasses and a face guard.
- * For best results, fine parameter adjustment may be required, depending on the materials; milling shape and strategy; machine rigidity and spindle capability.
- * Use a machine that has high rigidity and generates a low level of vibration.
- * Do not use flammable cutting oils.

Advisory for regrinding UNIMAX Tungsten Carbide End Mills

- * Never regrind the tool without wearing safety glasses and a face guard.



UNION TOOL CO.

U.S. UNION TOOL, INC.

(U.S. HEADQUARTERS)

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2962 Scott Boulevard, Santa Clara, CA 95054 U.S.A.

Tel: 1-408-982-0205 Fax: 1-408-982-0320

UPPER MIDWEST REGIONAL SERVICE CENTER

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155 Bridgepoint Drive, Unit 3 South St. Paul, MN 55075 U.S.A.

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